Experiment 1 – Linear Regression

1. R^2

$SSE = \Sigma(y_i - \hat{y}_i)^2$ $SST = \Sigma(y_i - \bar{y})^2$ $R^2 = 1 - SSE/SST$

2. Adjusted R^2

$R^2_{adj} = 1 - (1 - R^2)\cdot(n - 1)/(n - p - 1)$

3. t-statistic

$t = \beta / SE(\beta)$

4. 95% Confidence Interval

$CI = \beta \pm 1.96\ (SE)$

5. Logistic Regression – Odds Ratio

Feature	Coefficient	Odds Ratio	% change
Age	0.06	1.06	+6%
Income	0.85	2.34	+134%
Loan Amount	−0.40	0.67	−33%
Credit Score	1.25	3.49	+249%

Experiment 2 – Decision Tree and KNN

1. Gini Impurity (Dataset: 6 Yes, 4 No)

$Gini = 1 - (0.6)^2 - (0.4)^2 = 1 - 0.36 - 0.16 = 0.48$

2. Gini after split

Left: 4 Yes, 0 No → $Gini_L = 0$

Right: 2 Yes, 4 No → $Gini_R = 1 - (2/6)^2 - (4/6)^2 = 0.4444$

Weighted Gini = $(4/10)(0) + (6/10)(0.4444) = 0.2667$

3. Gini Gain

$= 0.48 - 0.2667 = 0.2133$

4. KNN Classification Query Q = (3,3)

Point	Coordinates	Distance
A	(1,1)	2.828
B	(2,2)	1.414
C	(4,4)	1.414
D	(5,5)	2.828

For K = 3, nearest neighbours: B, C, A (or D, depending on tie handling).

If classes are B = Yes, C = Yes, A = No: Prediction = Yes

5. KNN Regression

Nearest neighbour values: 20, 30, 40.

Prediction = (20 + 30 + 40)/3 = 30

Experiment 3 – Approximate Shapley Values

$f(x) = 2x_1 + 3x_2 + 5x_3$; $x = (4,2,3)$, $z = (5,1,4)$

Initial prediction: $f(z) = 2(5) + 3(1) + 5(4) = 33$

Final prediction: $f(x) = 2(4) + 3(2) + 5(3) = 29$

Feature contributions:

$\varphi_1 = 2(4 - 5) = -2$ $\varphi_2 = 3(2 - 1) = 3$ $\varphi_3 = 5(3 - 4) = -5$

Verification: $-2 + 3 - 5 = -4$

$f(x) - f(z) = 29 - 33 = -4$ ✓

Experiment 4 – PDP-Based Feature Importance

Model: $f(x) = 0.01(Age) + 0.0001(Income) + Category$

Dataset (Age, Income, Category): (20, 30000, A), (30, 40000, B), (40, 50000, A), (50, 60000, C)

Category values: A = 0.1, B = 0.2, C = 0.3.

1. PDP for Age

PDP = [4.875, 4.975, 5.075, 5.175]; Mean PDP = 5.025

$I = \sqrt{[(0.0225 + 0.0025 + 0.0025 + 0.0225)/3]}$ → I(Age) = 0.1291

2. PDP for Income

PDP = [3.525, 4.525, 5.525, 6.525]; Mean PDP = 5.025

$I = \sqrt{[(2.25 + 0.25 + 0.25 + 2.25)/3]}$ → I(Income) = 1.2910

3. PDP for Category

PDP(A) = 4.95, PDP(B) = 5.05, PDP(C) = 5.15

I = (5.15 − 4.95)/4 → I(Category) = 0.05

Ranking

1. Income = 1.2910 2. Age = 0.1291 3. Category = 0.0500

Experiment 5 – ICE Plot

$f(x_1,x_2) = 3.5 + 3x_1 + 4x_2$; X = [(5,2), (6,3), (7,4)]; Grid x_1 = [5, 6, 7]

Sample 1 (x_2=2): f(5,2) = 3.5+15+8 = 26.5; f(6,2) = 3.5+18+8 = 29.5; f(7,2) = 3.5+21+8 = 32.5

Sample 2 (x_2=3): f(5,3) = 30.5; f(6,3) = 33.5; f(7,3) = 36.5

Sample 3 (x_2=4): f(5,4) = 34.5; f(6,4) = 37.5; f(7,4) = 40.5

PDP

PDP(5) = (26.5 + 30.5 + 34.5)/3 = 30.5

PDP(6) = (29.5 + 33.5 + 37.5)/3 = 33.5

PDP(7) = (32.5 + 36.5 + 40.5)/3 = 36.5

Experiment 6 – XRAI

Attribution matrix A (rows): [0.1 0.2 0.1 0.0]; [0.2 0.3 0.2 0.1]; [0.0 0.1 0.8 0.7]; [0.1 0.2 0.9 0.8]

R1 = (0.1 + 0.2 + 0.2 + 0.3)/4 = 0.20

R2 = (0.1 + 0.0 + 0.2 + 0.1)/4 = 0.10

R3 = (0.0 + 0.1 + 0.1 + 0.2)/4 = 0.10

R4 = (0.8 + 0.7 + 0.9 + 0.8)/4 = 0.80

Ranking: 1. R4 = 0.80 2. R1 = 0.20 3. R2 = 0.10 4. R3 = 0.10

Most Important Region = R4

Experiment 7 – Grad-CAM

A_1 = [1 2; 3 4], A_2 = [2 1; 4 3] G_1 = [0.1 0.2; 0.3 0.4], G_2 = [0.2 0.1; 0.2 0.3]

1. Calculate weights
$Z = 2 \times 2 = 4$
$\alpha_1 = (0.1+0.2+0.3+0.4)/4 = 0.25$ $\alpha_2 = (0.2+0.1+0.2+0.3)/4 = 0.20$

2. Multiply feature maps by weights
$0.25 \cdot A_1 = [0.25\ 0.50; 0.75\ 1.00]$ $0.20 \cdot A_2 = [0.40\ 0.20; 0.80\ 0.60]$

3. Add weighted feature maps
$M = [0.25+0.40\ \ 0.50+0.20; 0.75+0.80\ \ 1.00+0.60] = [0.65\ 0.70; 1.55\ 1.60]$

4. Apply ReLU $ReLU(x) = \max(0, x)$
$L^{Grad-CAM} = [0.65\ 0.70; 1.55\ 1.60]$

5. Normalize heatmap (max = 1.60, $L_{norm} = L/1.60$)
$[0.406\ 0.438; 0.969\ 1.000]$
Most important region: Bottom-right pixel (value = 1.000).

Experiment 8 – LIME Using Weighted Least Squares

1. Given data
Original x = [1,1]; perturbed $z_1 = [1,1]$, $z_2 = [1,0]$, $z_3 = [0,1]$, $z_4 = [0,0]$
Black-box predictions y = [0.90, 0.70, 0.60, 0.20]; kernel width $\sigma = 1$

2. Euclidean distances $d(x,z) = \sqrt{[(x_1-z_1)^2 + (x_2-z_2)^2]}$
$d_1 = \sqrt{(0+0)} = 0$ $d_2 = \sqrt{(0+1)} = 1$ $d_3 = \sqrt{(1+0)} = 1$ $d_4 = \sqrt{(1+1)} = 1.414$

3. Similarity weights $w_i = e^{-d_i^2/\sigma^2}$
$w_1 = e^0 = 1$ $w_2 = e^{-1} = 0.368$ $w_3 = e^{-1} = 0.368$ $w_4 = e^{-2} = 0.135$

4. Design matrix X (rows)
[1 1 1]; [1 1 0]; [1 0 1]; [1 0 0]

5. Weight matrix W
diag(1, 0.368, 0.368, 0.135)

6. Prediction matrix y
[0.90; 0.70; 0.60; 0.20]

7. X^TWX
[1.871 1.368 1.368; 1.368 1.368 0.368; 1.368 0.368 1.368]

8. X^TWy
[1.600; 1.158; 1.079]

9. Coefficients $\beta = (X^TWX)^{-1}X^TWy$
Solving the system $(X^TWX)\beta = X^TWy$:
$\beta_0 = 0.5712$ $\beta_1 = 0.2337$ $\beta_2 = 0.1547$

10. Final interpretation
Blood Pressure: +0.2337 (Positive); Cholesterol: +0.1547 (Positive)
$\beta_1 > \beta_2 > 0$ → Blood pressure has a larger positive local contribution than cholesterol.

11. Test Case – Negative Feature Contribution
(X1, X2, Prediction): (1,1,0.80), (1,0,0.90), (0,1,0.50), (0,0,0.40)
Effect of X1: 0.90 – 0.50 = +0.40 Effect of X2: 0.80 – 0.90 = –0.10
Therefore X1 = Positive contribution, X2 = Negative contribution

12. Test Case – Different Kernel Widths $w = e^{-d^2/\sigma^2}$
Distances: 0, 1, 1, 1.414
Distance | σ=0.5 | σ=1 | σ=2
0 | 1.000 | 1.000 | 1.000
1 | 0.018 | 0.368 | 0.779
1 | 0.018 | 0.368 | 0.779
1.414 | 0.0003 | 0.135 | 0.607

Experiment 9 – CBOW Word Embeddings and Gradient × Input

Part A – CBOW

1. Tokenization: sentences "the movie was good", "the movie was bad"
$S_1 = [the, movie, was, good]$; $S_2 = [the, movie, was, bad]$
2. Vocabulary: V = [the, movie, was, good, bad], |V| = 5
3. One-hot: the = [1,0,0,0,0]; movie = [0,1,0,0,0]; was = [0,0,1,0,0]; good = [0,0,0,1,0]; bad = [0,0,0,0,1]
4. CBOW training examples (window = 1): (the, was) → movie; (movie, good) → was; (the, was) → movie; (movie, bad) → was
5. Matrix dimensions: vocab = 5, embedding dim = 2 → W = 5×2, W′ = 2×5
6. Average context embedding: $e_{movie} = [0.2, 0.4]$, $e_{was} = [0.6, 0.8]$
$h = (e_{movie} + e_{was})/2 = [(0.2+0.6)/2, (0.4+0.8)/2] = h = [0.4, 0.6]$
7. Output scores. W′ = [0.1 0.2 0.3 0.4 0.5; 0.2 0.1 0.2 0.3 0.4]; $z_j = h_1 W'_{1j} + h_2 W'_{2j}$
$z_1 = (0.4)(0.1)+(0.6)(0.2) = 0.16$; $z_2 = (0.4)(0.2)+(0.6)(0.1) = 0.14$; $z_3 = (0.4)(0.3)+(0.6)(0.2) = 0.24$
$z_4 = (0.4)(0.4)+(0.6)(0.3) = 0.34$; $z_5 = (0.4)(0.5)+(0.6)(0.4) = 0.44$
$z = [0.16, 0.14, 0.24, 0.34, 0.44]$
8. Softmax $P_j = e^{z_j} / \Sigma e^{z_k}$: P = [0.181, 0.178, 0.196, 0.217, 0.228]
9. Cross-entropy (target = movie, P(movie) = 0.178): L = –log(P(movie)) = –log(0.178) ≈ 1.726
10. Gradient: ŷ = [0.10, 0.20, 0.15, 0.45, 0.10], target good y = [0,0,0,1,0]
$\partial L/\partial z_j = \hat{y}_j - y_j$ → [0.10, 0.20, 0.15, –0.55, 0.10]
11. Output matrix update: $W'_{new} = W'_{old} - \eta(h)(\hat{y} - y)$, $\eta = 0.05$

$W'_{11,new} = 0.1 - (0.05)(0.4)(0.10) = 0.098$

12. Embedding update: $W'(\hat{y} - y) = [-0.02; -0.12] = \partial L/\partial h$. For context word movie:

$e_{movie,new} = e_{movie} - (0.05/2)[-0.02; -0.12] = [0.2, 0.4] - [-0.0005, -0.003] = [0.2005, 0.403]$

Note: The CBOW code trains for 1,000 epochs with random initialization. Final embeddings and cosine similarities depend on the training updates; the manual does not specify one fixed final numerical answer.

Part B – Gradient × Input

$w = [0.1, 0.2, 0.1, 2, -2]$, $b = 0$ (order: the, movie, was, good, bad)

1. Positive sentence "the movie was good": $x = [1,1,1,1,0]$

$z = w^T x + b = (0.1)(1)+(0.2)(1)+(0.1)(1)+(2)(1)+(-2)(0) = 2.4$; $\hat{y} = 1/(1+e^{-2.4}) = 0.9168$

2. Sigmoid gradient: $\hat{y}(1 - \hat{y}) = 0.9168(1 - 0.9168) = 0.0763$

3. Gradient$_i$ = 0.0763·w_i: the 0.0763(0.1) = 0.00763; movie 0.0763(0.2) = 0.01526; was 0.0763(0.1) = 0.00763;

good 0.0763(2) = 0.1526; bad 0.0763(−2) = −0.1526

4. Gradient × Input, $A_i = x_i \times$ Gradient$_i$: the 1×0.00763 = 0.00763; movie 1×0.01526 = 0.01526;

was 1×0.00763 = 0.00763; good 1×0.1526 = 0.1526; bad 0×(−0.1526) = 0. Most important word = good

5. Negative sentence "the movie was bad": $x = [1,1,1,0,1]$

$z = 0.1 + 0.2 + 0.1 + (2)(0) + (-2)(1) = -1.6$; $\hat{y} = 1/(1+e^{1.6}) = 0.1680$

Sigmoid gradient: 0.1680(1 − 0.1680) = 0.1398

6. Gradient × Input: the 1, grad 0.01398 → +0.01398; movie 1, 0.02796 → +0.02796; was 1, 0.01398 → +0.01398;

good 0, +0.2796 → 0; bad 1, −0.2796 → −0.2796. Most important word = bad

Experiment 10 – Layer-wise Relevance Propagation (LRP)

Weights: $w_{11} = 1, w_{21} = 0, w_{12} = 0, w_{22} = 2$; $v_1 = 1, v_2 = 1$

1. Forward propagation

$h_1 = \text{ReLU}(x_1 w_{11} + x_2 w_{21})$; $h_2 = \text{ReLU}(x_1 w_{12} + x_2 w_{22})$; $s = h_1 v_1 + h_2 v_2$; $P = 1/(1+e^{-s})$

2. Relevance propagation

$R_{output} = s$; $R_{h1} = [h_1 v_1/(h_1 v_1 + h_2 v_2)]\cdot R_{output}$; $R_{h2} = [h_2 v_2/(h_1 v_1 + h_2 v_2)]\cdot R_{output}$

$R_{x1} = (x_1 w_{11}/d_1)R_{h1} + (x_1 w_{12}/d_2)R_{h2}$; $R_{x2} = (x_2 w_{21}/d_1)R_{h1} + (x_2 w_{22}/d_2)R_{h2}$

where $d_1 = x_1 w_{11} + x_2 w_{21}$, $d_2 = x_1 w_{12} + x_2 w_{22}$

Test Case 1: $x_1 = 1, x_2 = 1$

$h_1 = \text{ReLU}(1(1)+1(0)) = 1$; $h_2 = \text{ReLU}(1(0)+1(2)) = 2$; $s = 1(1)+2(1) = 3$; $P = 1/(1+e^{-3}) = 0.9526$

$R_{output} = 3$; $R_{h1} = (1/3)(3) = 1$; $R_{h2} = (2/3)(3) = 2$; $R_{x1} = (1/1)(1) + 0 = 1$; $R_{x2} = 0 + (2/2)(2) = 2$

Conservation: 1 + 2 = 3

Test Case 2: $x_1 = 2, x_2 = 1$

$h_1 = 2, h_2 = 2$; $s = 2+2 = 4$; $P = 1/(1+e^{-4}) = 0.9820$

$R_{h1} = (2/4)(4) = 2$; $R_{h2} = (2/4)(4) = 2$; $R_{x1} = 2$; $R_{x2} = 2$; Conservation: 2 + 2 = 4

Test Case 3: $x_1 = 0, x_2 = 1$

$h_1 = 0, h_2 = 2$; $s = 0+2 = 2$; $P = 1/(1+e^{-2}) = 0.8808$

$R_{h1} = 0$; $R_{h2} = 2$; $R_{x1} = 0$; $R_{x2} = 2$; Conservation: 0 + 2 = 2

Test Case 4: $x_1 = 1, x_2 = 0$

$h_1 = 1, h_2 = 0$; $s = 1+0 = 1$; $P = 1/(1+e^{-1}) = 0.7311$

$R_{h1} = 1$; $R_{h2} = 0$; $R_{x1} = 1$; $R_{x2} = 0$; Conservation: 1 + 0 = 1

Test Case 5: $x_1 = 0, x_2 = 0$

$h_1 = 0, h_2 = 0$; $s = 0$; $P = 1/(1+e^{0}) = 0.5$

Denominator is zero, so the program assigns all relevance values zero: $R_{h1} = R_{h2} = R_{x1} = R_{x2} = 0$

Final answers

Case	Input	Hidden act.	Logit	Prob.	Input relevance
1	(1,1)	(1,2)	3	0.9526	(1,2)
2	(2,1)	(2,2)	4	0.9820	(2,2)
3	(0,1)	(0,2)	2	0.8808	(0,2)
4	(1,0)	(1,0)	1	0.7311	(1,0)
5	(0,0)	(0,0)	0	0.5000	(0,0)

For the first four cases: $R_{x1} + R_{x2} = R_{output}$